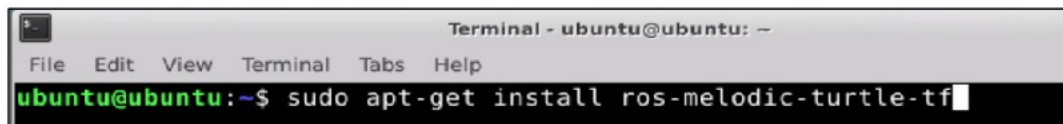


第 13 课 TF 坐标系广播与监听的编程实现

1.机器人的坐标变换

在进行编程前，先需要了解机器人的坐标变换。这里以运行海龟案例来进行解说，步骤如下：

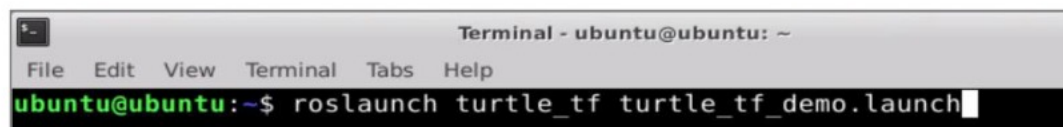
- 1) 输入指令“`sudo apt-get install ros-melodic-turtle-tf`”，安装功能包。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ sudo apt-get install ros-melodic-turtle-tf
```

其中“**melodic**”对应的是ROS版本。

- 2) 输入指令“`roslaunch turtle_tf turtle_tf_demo.launch`”，启动launch文件。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roslaunch turtle_tf turtle_tf_demo.launch
```



- 3) 输入指令“**roslaunch turtlesim turtle_teleop_key**”，运行海龟键盘控制节点。

```
ubuntu@ubuntu:~$ roslaunch turtlesim turtle_teleop_key
Reading from keyboard
-----
```

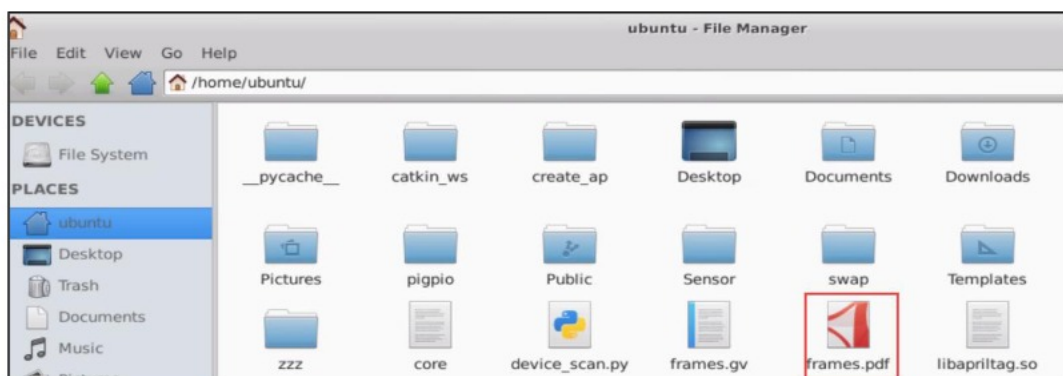


- 4) 输入指令“**roslaunch tf view_frames**”，可视化框架。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roslaunch tf view_frames
the rosdep view is empty: call 'sudo rosdep init' and 'rosdep update'
Listening to /tf for 5.0 seconds
Done Listening
dot - graphviz version 2.40.1 (20161225.0304)

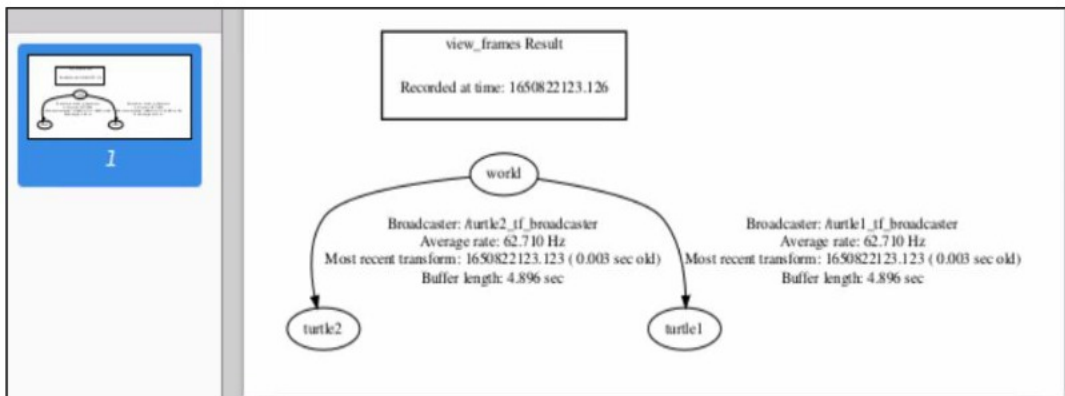
Detected dot version 2.40
frames.pdf generated
```

在主目录下可以找到“**frames.pdf**”文件，如下图：





打开“frames.pdf”文件，可以看到当前系统TF坐标之间的位置关系，如下图：



2.创建功能包

下面，就开始创建功能包，步骤如下：

- 1) 输入指令“`cd catkin_ws/src/`”，回车，进入工作空间。

```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ cd catkin_ws/src/
```

- 5) 输入指令“`catkin_create_pkg tf_hiwonder rospy std_msgs`”，回车，创建功能包。

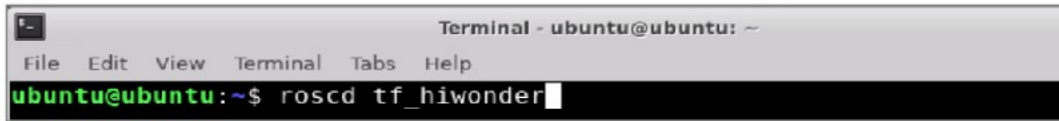
```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src$ catkin_create_pkg tf_hiwonder rospy std_msgs
```

3.编程方法

3.1 编写广播和监听程序

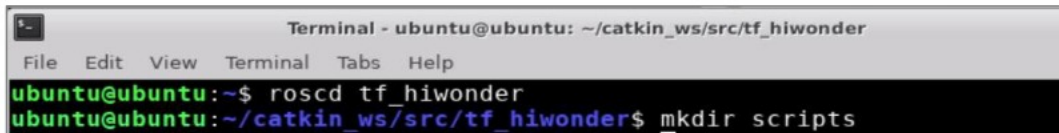
- 1) 打开Linux命令行终端。

- 2) 输入进入功能包所在目录的指令“**roscd tf_hiwonder**”，并按下回车。



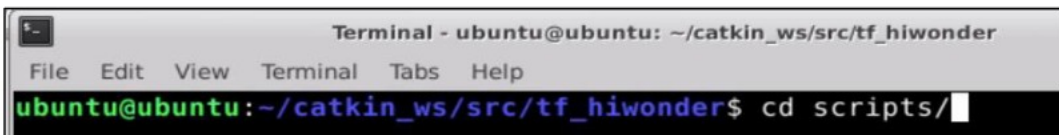
```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roscd tf_hiwonder
```

- 3) 输入指令“**mkdir scripts**”，并按下回车，新建用于存放Python脚本的文件夹“**scripts**”。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roscd tf_hiwonder
ubuntu@ubuntu:~/catkin_ws/src/tf_hiwonder$ mkdir scripts
```

- 4) 输入指令“**cd scripts/**”，并按下回车，进入用于存放Python脚本的文件夹“**scripts**”。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/tf_hiwonder$ cd scripts/
```

- 5) 输入指令“**vi turtle_tf_broadcaster.py**”编辑程序，复制下面程序。如需修改，再按下“**i**”即可修改。修改完成，按下“**Esc**”，输入“**: wq**”保存并退出。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/tf_hiwonder/scripts$ vi turtle_tf_broadcaster.py
```

```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

# 该例程将请求/show_person服务，服务数据类型learning_service::Person

import tf

import rospy

import turtlesim.msg

def handle_turtle_pose(msg, turtlename):

    br = tf.TransformBroadcaster()
```

```

br.sendTransform((msg.x, msg.y, 0),

                 tf.transformations.quaternion_from_euler(0, 0, msg.theta),

                 rospy.Time.now(),

                 turtlename,

                 "world")

if __name__ == '__main__':

    rospy.init_node('turtle_tf_broadcaster')

    turtlename = rospy.get_param('~turtle')

    rospy.Subscriber('/%s/pose' % turtlename,

                     turtlesim.msg.Pose,

                     handle_turtle_pose,

                     turtlename)

    rospy.spin()

```

```

Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder/scripts
File Edit View Terminal Tabs Help
2 # -*- coding: utf-8 -*-
3 # 该例程将请求/show_person服务，服务数据类型 learning_service::Person
4 import tf
5 import rospy
6 import turtlesim.msg
7
8 def handle_turtle_pose(msg, turtlename):
9     br = tf.TransformBroadcaster()
10     br.sendTransform((msg.x, msg.y, 0),
11                     tf.transformations.quaternion_from_euler(0, 0, msg.theta),
12                     rospy.Time.now(),
13                     turtlename,
14                     "world")
15
16 if __name__ == '__main__':
17     rospy.init_node('turtle_tf_broadcaster')
18     turtlename = rospy.get_param('~turtle')
19     rospy.Subscriber('/%s/pose' % turtlename,
20                     turtlesim.msg.Pose,
21                     handle_turtle_pose,
22                     turtlename)
23     rospy.spin()
:wq

```

6) 输入指令“**vi turtle_tf_listener.py**”编辑程序，复制下面程序。如需修改，再按下“**i**”即可修改。修改完成，按下“**Esc**”，输入“**: wq**”保存并退出。



```
#!/usr/bin/env python

# -*- coding: utf-8 -*-

# 该例程将请求/show_person服务，服务数据类型learning_service::Person

import roslib

import rospy

import math

import tf

import geometry_msgs.msg

import turtlesim.srv

if __name__ == '__main__':

    rospy.init_node('turtle_tf_listener')

    listener = tf.TransformListener()

    rospy.wait_for_service('spawn')

    spawner = rospy.ServiceProxy('spawn', turtlesim.srv.Spawn)
```



```
spawner(4, 2, 0, 'turtle2')
```

```
turtle_vel = rospy.Publisher('turtle2/cmd_vel', geometry_msgs.msg.Twist, queue_size=1)
```

```
rate = rospy.Rate(10.0)
```

```
while not rospy.is_shutdown():
```

```
    try:
```

```
        (trans, rot) = listener.lookupTransform('/turtle2', '/turtle1', rospy.Time(0))
```

```
    except (tf.LookupException, tf.ConnectivityException, tf.ExtrapolationException):
```

```
        continue
```

```
    angular = 4 * math.atan2(trans[1], trans[0])
```

```
    linear = 0.5 * math.sqrt(trans[0]**2 + trans[1]**2)
```

```
    cmd = geometry_msgs.msg.Twist()
```

```
    cmd.linear.x = linear
```

```
    cmd.angular.z = angular
```

```
    turtle_vel.publish(cmd)
```

```
    rate.sleep()
```

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder/scripts
File Edit View Terminal Tabs Help
18  spawner = rospy.ServiceProxy('spawn', turtlesim.srv.Spawn)
19  spawner(4, 2, 0, 'turtle2')
20
21  turtle_vel = rospy.Publisher('turtle2/cmd_vel', geometry_msgs.msg.Twist,
queue_size=1)
22
23  rate = rospy.Rate(10.0)
24  while not rospy.is_shutdown():
25      try:
26          (trans,rot) = listener.lookupTransform('/turtle2', '/turtle1', r
rospy.Time(0))
27      except (tf.LookupException, tf.ConnectivityException, tf.Extrapolati
onException):
28          continue
29
30      angular = 4 * math.atan2(trans[1], trans[0])
31      linear = 0.5 * math.sqrt(trans[0] ** 2 + trans[1] ** 2)
32      cmd = geometry_msgs.msg.Twist()
33      cmd.linear.x = linear
34      cmd.angular.z = angular
35      turtle_vel.publish(cmd)
36
37      rate.sleep()
:wq
```

6) 输入指令“`chmod +x turtle_tf_broadcaster.py`”和“`chmod +x turtle_tf_listener.py`”，并按下回车，赋予文件可执行权限。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/tf_hiwonder/scripts$ chmod +x turtle_tf_broadcaster.py
```

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/tf_hiwonder/scripts$ chmod +x turtle_tf_listener.py
```

7) 输入指令“`cd ..`”和“`mkdir launch`”，新建用于存放launch脚本的文件夹“`launch`”。

```
Terminal - ubuntu@ubuntu: ~/Desktop/tf_hiwonder/scripts
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/Desktop/tf_hiwonder/scripts$ cd ..
```

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/tf_hiwonder$ mkdir launch
```

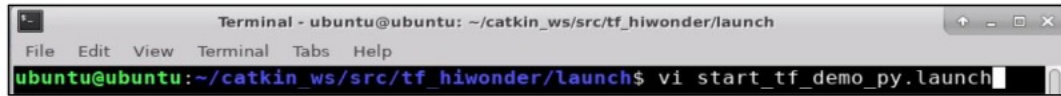
8) 输入指令“`cd launch/`”，并按下回车，进入用于存放Python脚本的文件夹“`launch`”。

```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws/src/tf_hiwonder$ cd launch/
```

9) 输入指令“`vi start_tf_demo_py.launch`”编辑程序，复制下面程序。如需修改，再



按下“**i**”即可修改。修改完成，按下“**Esc**”，输入“**: wq**”保存并退出。



```
<launch>

  <!-- Turtlesim Node-->

  <node pkg="turtlesim" type="turtlesim_node" name="sim"/>

  <node pkg="turtlesim" type="turtle_teleop_key" name="teleop" output="screen"/>


  <node          name="turtle1_tf_broadcaster"          pkg="tf_hiwonder"
type="turtle_tf_broadcaster.py" respawn="false" output="screen" >

    <param name="turtle" type="string" value="turtle1" />

  </node>

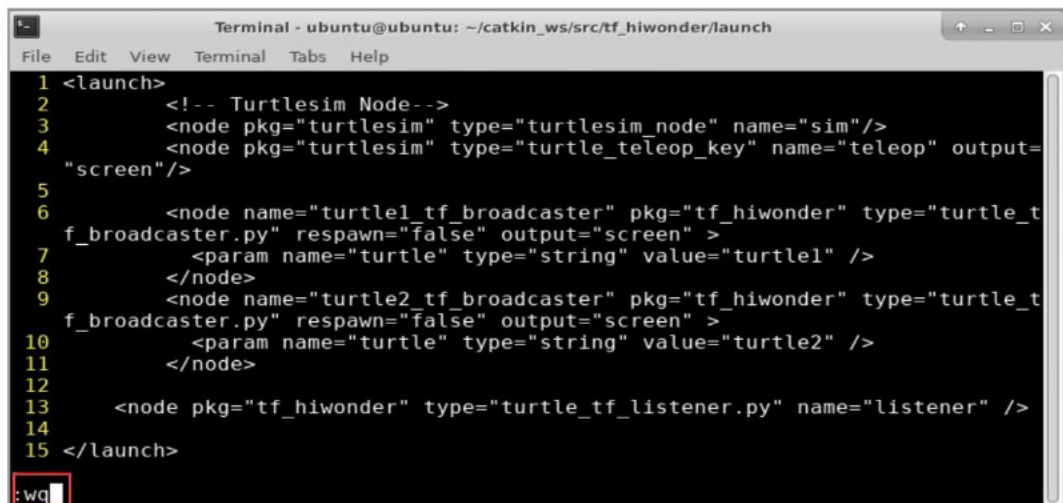
  <node          name="turtle2_tf_broadcaster"          pkg="tf_hiwonder"
type="turtle_tf_broadcaster.py" respawn="false" output="screen" >

    <param name="turtle" type="string" value="turtle2" />

  </node>


  <node pkg="tf_hiwonder" type="turtle_tf_listener.py" name="listener" />

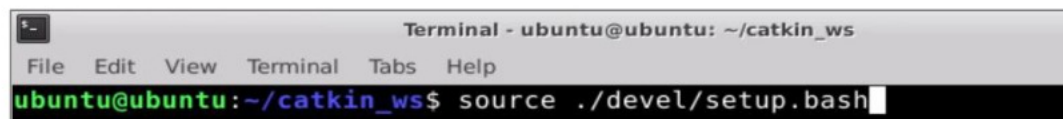

</launch>
```



```
Terminal - ubuntu@ubuntu: ~/catkin_ws/src/tf_hiwonder/launch
1 <launch>
2   <!-- Turtlesim Node-->
3   <node pkg="turtlesim" type="turtlesim_node" name="sim"/>
4   <node pkg="turtlesim" type="turtle_teleop_key" name="teleop" output=
"screen"/>
5
6   <node name="turtle1_tf_broadcaster" pkg="tf_hiwonder" type="turtle_t
f_broadcaster.py" respawn="false" output="screen" >
7     <param name="turtle" type="string" value="turtle1" />
8   </node>
9   <node name="turtle2_tf_broadcaster" pkg="tf_hiwonder" type="turtle_t
f_broadcaster.py" respawn="false" output="screen" >
10    <param name="turtle" type="string" value="turtle2" />
11  </node>
12
13  <node pkg="tf_hiwonder" type="turtle_tf_listener.py" name="listener" />
14
15 </launch>
:wq
```

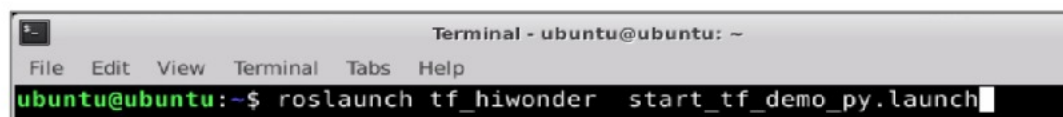
3.2 运行程序

- 1) 输入指令“`source ./devel/setup.bash`”，并按下回车，设置工作环境。



```
Terminal - ubuntu@ubuntu: ~/catkin_ws
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~/catkin_ws$ source ./devel/setup.bash
```

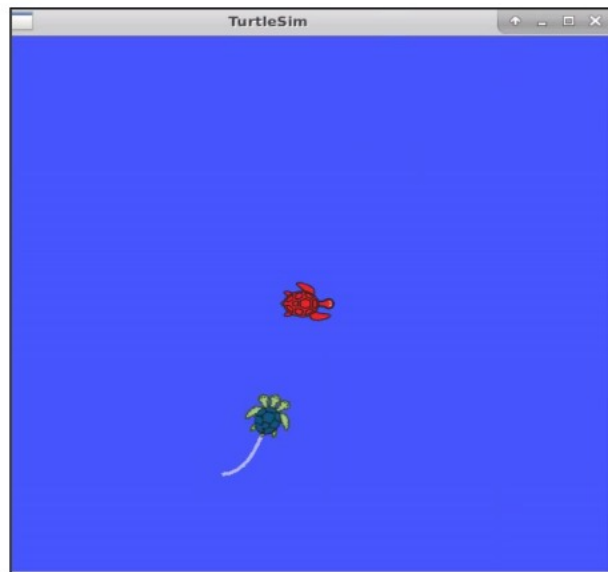
- 2) 输入指令“`roslaunch tf_hiwonder start_tf_demo_py.launch`”，并按下回车，运行launch程序。



```
Terminal - ubuntu@ubuntu: ~
File Edit View Terminal Tabs Help
ubuntu@ubuntu:~$ roslaunch tf_hiwonder start_tf_demo_py.launch
```



一只海龟自动行走到另一只海龟位置，显示如下图：



3) 如需程序，可按下快捷键“**Ctrl+C**”。